Team -Code Max

PROBLEM SPACE :- HEALTH CARE

Basic SRS / Implementation Plan: CareLoop

1. System Overview & Technical Architecture

CareLoop is a full-stack MERN (MongoDB, Express.js, React.js, Node.js) web application designed for high-volume district hospital outpatient departments (~900 visits/day). It targets outpatient loss-to-follow-up by chaining four deterministic modules: C1 (Follow-Up Structurer), C2 (Shared-Phone Contact Resolver), C3 (Contact Consent Ledger), and C4 (CHW Handoff List).

• Frontend: React 18 (Vite), Tailwind CSS.

• Backend: Node.js + Express.js REST API with JWT RBAC.

• Database: MongoDB Atlas (enforcing immutable audit timestamps).

• AI Layer: Google Gemini 2.5 Flash API using strict responseSchema JSON mode.

2. Users & Role Definitions

• Registration & Records Clerk: Captures phone ownership (Personal/Shared/Unreachable), logs alternate contact windows, and records explicit patient consent at registration.

• OPD Clinician (Doctor / Nurse): Enters rapid free-text clinical follow-up note, reviews the AI-generated structured card, assigns priority, and confirms.

• Community Health Worker (CHW): Accesses the village-clustered handoff list of the 7% unreachable (and consented) patients to conduct home visits.

• Patient / Caregiver: Receives the bilingual structured follow-up card at checkout and via consented contact channels.

3. System Inputs & Outputs

Inputs:

• Registration: patient_id, full_name, age, phone_status, shared_phone_metadata, preferred_contact_window.

• Consent: consent_caregiver_relay, consent_chw_home_visit, privacy_masking_level.

• Clinical: raw_clinical_note (Free-text string), clinician_priority_tag.

Outputs:

• Confirmed Structured Card (JSON): Contains target_date, window_days, why_return, what_to_bring[], and doctor_confirmed boolean.

• Consented Outreach Payload: Scheduled message queue filtered by preferred window and privacy level.

• CHW Handoff List: Village/ward-grouped CSV/PDF for unreachable/missed-window patients.

4. End-to-End Operational Workflow

1. Intake (C2): Clerk records demographics and selects phone_status.

2. Consent (C3): Clerk logs consent for caregiver relay and CHW visits.

3. AI Structuring (C1): Doctor types free-text note; AI structures When/Why/Bring fields.

4. Human Confirmation (C1): Doctor reviews the preview and clicks 'Confirm & Save'.

5. Routing (C4): System routes reminders via personal/shared phone, or adds unreachable/consented patients directly to the localized CHW Handoff List.

5. Functional Requirements

• FR-C1.1 (Parsing): Parse unstructured clinical notes into three mandatory structured fields within 2.0 seconds.

• FR-C1.2 (Ambiguity): If a timeframe is missing, flag it and disable the save button until the clinician manually selects a date.

• FR-C1.3 (Confirmation Lock): Reject database persistence unless doctor_confirmed == true.

• FR-C3.1 (Audit): Modifications to consent status must append timestamped audit entries, never overwrite.

• FR-C4.1 (Automated Handoff): Unreachable patients with CHW consent must automatically populate the CHW Handoff List.

6. AI vs. Human Responsibilities

• AI Responsibilities: Extract target return date, reason, and checklist from shorthand. Translate checklists into local language. Enforce system validation rules.

• Human Responsibilities: Clinicians maintain authoritative accountability by reviewing, editing, assigning priority, and signing the card. Clerks secure real-world consent.

7. System Constraints

• Zero Autonomous Clinical Action: The system must never generate new medical advice, alter medication, or diagnose.

• Throughput: C1 review-and-confirm must take < 5 seconds of clinician time.

• Privacy: For shared phones with GENERIC_REMINDER_ONLY, clinical reasons must be stripped from outbound messages.

8. Failure & Edge-Case Handling

• Missing Date: AI detects missing info; UI highlights date picker and disables save until human input.

• Conflicting Dates (e.g., '1 week / 1 month'): System detects conflict and forces clinician to select the intended date.

• Consent Denied for Unreachable: System blocks CHW handoff, prints physical slip only, and logs as SELF_MANAGED.

• API Timeout: UI gracefully falls back to a deterministic regex/manual quick-select form to preserve clinic flow.

9. Acceptance Criteria & Metrics

• AT-01 (Missing-Date Safeguard): Given a note with no date, the system extracts instructions but blocks saving until the doctor provides a date.

• AT-02 (Consent Gate): Given a shared phone with caregiver relay denied, the system suppresses outbound messages and prints physical slip only.

• AT-03 (Unreachable Routing): Given an unreachable status + CHW consent, confirming the card immediately populates the local CHW list.

• Primary Metric: >95% of flagged visits exit with a confirmed structured card and a verified delivery pathway.

• Harm Metric: 0.0% of reminders sent without explicit ledger consent.

10. Version-1 Exclusions

• No autonomous clinical triage or medical diagnosis.

• No offline-first mobile app for CHWs (V1 utilizes a responsive web dashboard).

• No live telecom billing/IVR (V1 utilizes a simulated dispatch console for demonstration).

• No HL7-FHIR migration (V1 is a standalone MERN continuity layer).